SECTION 1

Voltaic Cells

TURNING CHEMICAL ENERGY INTO ELECTRICAL ENERGY

Standards

Electrochemistry: students explain how redox reactions convert chemical energy to electrical energy and back, and apply this to cells, batteries, corrosion, and electrolysis.

Objectives

- How does a voltaic cell convert chemical energy into electrical energy?
- What are the parts of a voltaic cell, and what does each part do?
- How is the standard potential of a cell calculated, and what does its sign tell us?

Expectations

- Identify the anode, cathode, half-cells, and salt bridge of a voltaic cell.
- Write half-reactions, the overall cell reaction, and cell notation.
- Calculate E°cell from standard reduction potentials and predict whether a reaction is spontaneous.

Section 1

Voltaic Cells

MAIN IDEA

In voltaic cells, oxidation takes place at the anode, yielding electrons that flow to the cathode, where reduction occurs.

KWS

WHAT I KNOW	WHAT I WANT TO KNOW	SOURCE
		YouTube Tyler DeWitt: Galvanic Cells (Voltaic Cells) SIMULATION AACT simulation: Galvanic/Voltaic Cells — build cells and read the voltmeter

Vocabulary

Review

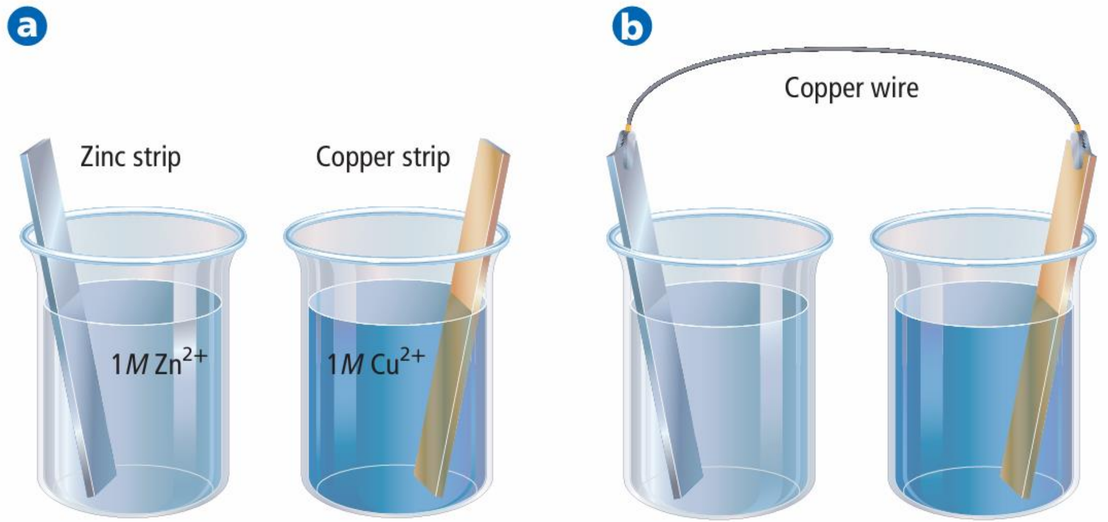
- **oxidation** — loss of electrons
- **reduction** — gain of electrons
- **redox reaction** — electrons are transferred from one species to another

New

- **salt bridge** — pathway that lets ions pass from one side to the other
- **electrochemical cell** — uses a redox reaction to produce electrical energy, or electrical energy to cause a reaction
- **voltaic cell** — converts chemical energy to electrical energy by a spontaneous redox reaction
- **half-cell** — one of the two parts of a cell, where oxidation or reduction occurs
- **anode** — electrode where oxidation takes place
- **cathode** — electrode where reduction occurs
- **reduction potential** — tendency of a substance to gain electrons
- **standard hydrogen electrode** — the reference against which reduction potentials are measured

Redox in Electrochemistry

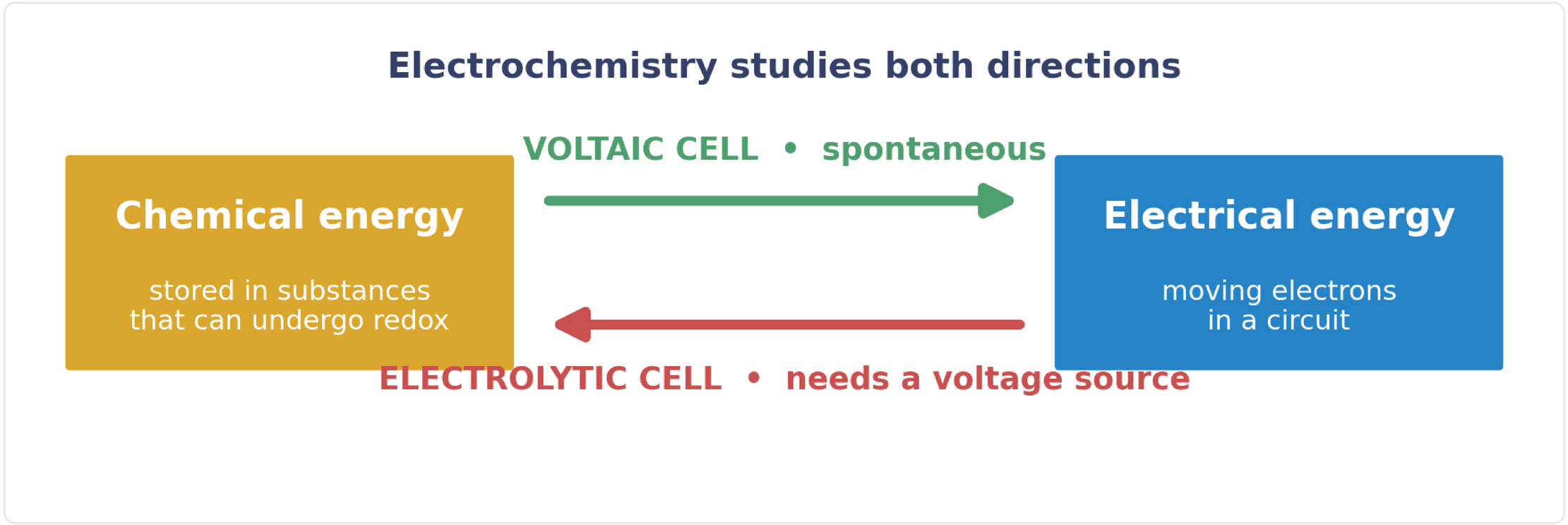
- **Electrochemistry** is the study of the redox processes by which chemical energy is converted to electrical energy and vice versa.
- Redox reactions involve a transfer of electrons from the species that is oxidized to the species that is reduced.
- If the zinc and copper half-reactions are separated, the electrons must travel through a wire — but a wire alone is not enough.



(a) separate half-cells (b) joined by a wire only — no sustained current

Two Directions of Energy Conversion

- The same redox chemistry can run in either direction — this unit studies both.



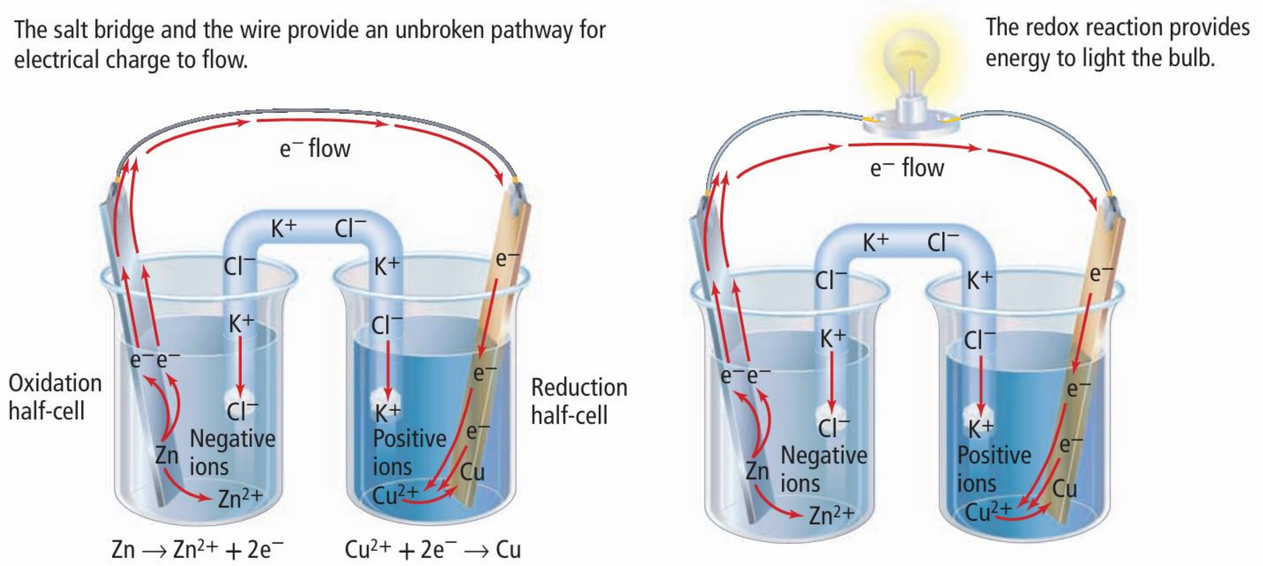
WATCH Tyler DeWitt: Introduction to Electrochemistry

Completing the Circuit: The Salt Bridge

- **A salt bridge** is a pathway to allow the passage of ions from one side to another, so that ions do not build up around the electrodes.

- **An electrochemical cell** is an apparatus that uses a redox reaction to produce electrical energy or uses electrical energy to cause a chemical reaction.

- **A voltaic cell** is a type of electrochemical cell that converts chemical energy to electrical energy by a spontaneous redox reaction.



Parts of a Voltaic Cell

- An electrochemical cell consists of two parts called half-cells, in which the separate oxidation and reduction reactions take place.
- The electrode where oxidation takes place is called the anode.
- **The cathode** is the electrode where reduction occurs.

Zinc–copper cell

ANODE (OXIDATION)

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

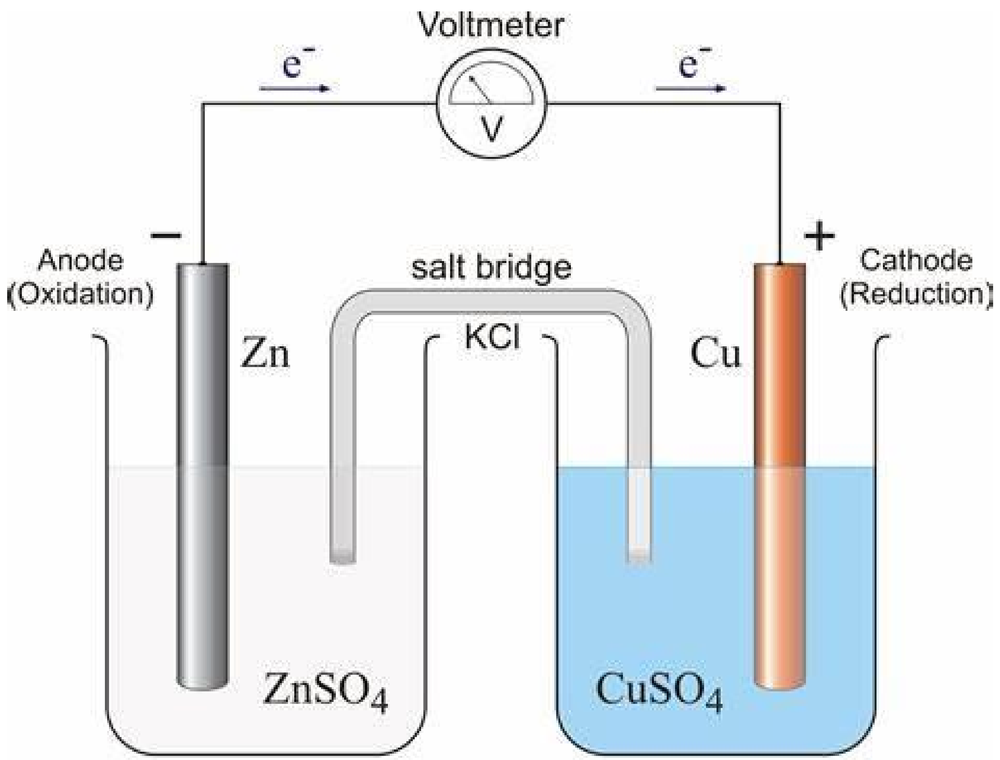
CATHODE (REDUCTION)

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

CELL NOTATION

$$Zn(s) \mid ZnSO_4(aq) \parallel CuSO_4(aq) \mid Cu(s)$$

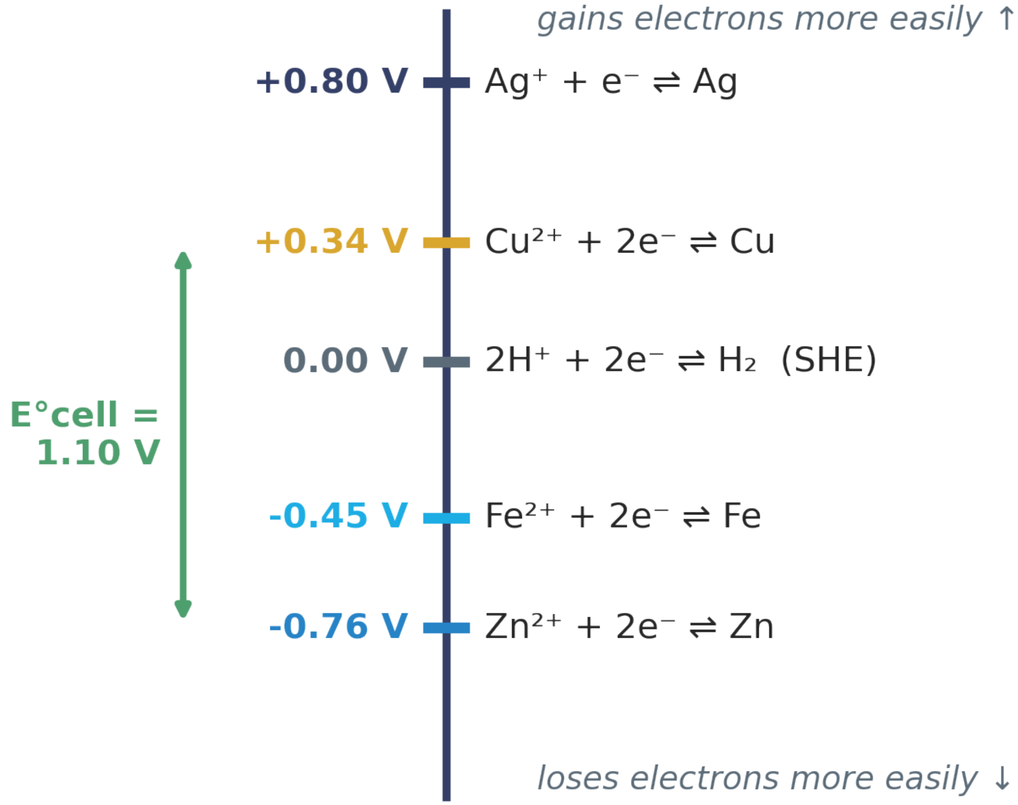
WATCH Tyler DeWitt: Galvanic Cells (Voltaic Cells)



Zinc–copper voltaic cell

Electric Potential and the Volt

- **Electric potential energy** is a measure of the amount of current that can be generated from a voltaic cell to do work.
- Electric charge can flow between two points only when a difference in electric potential energy exists between the two points.
- **A volt** is a unit used to measure cell potential — the force from the difference in electric potential energy between two electrodes.



The bigger the gap between two half-cells, the bigger the cell voltage

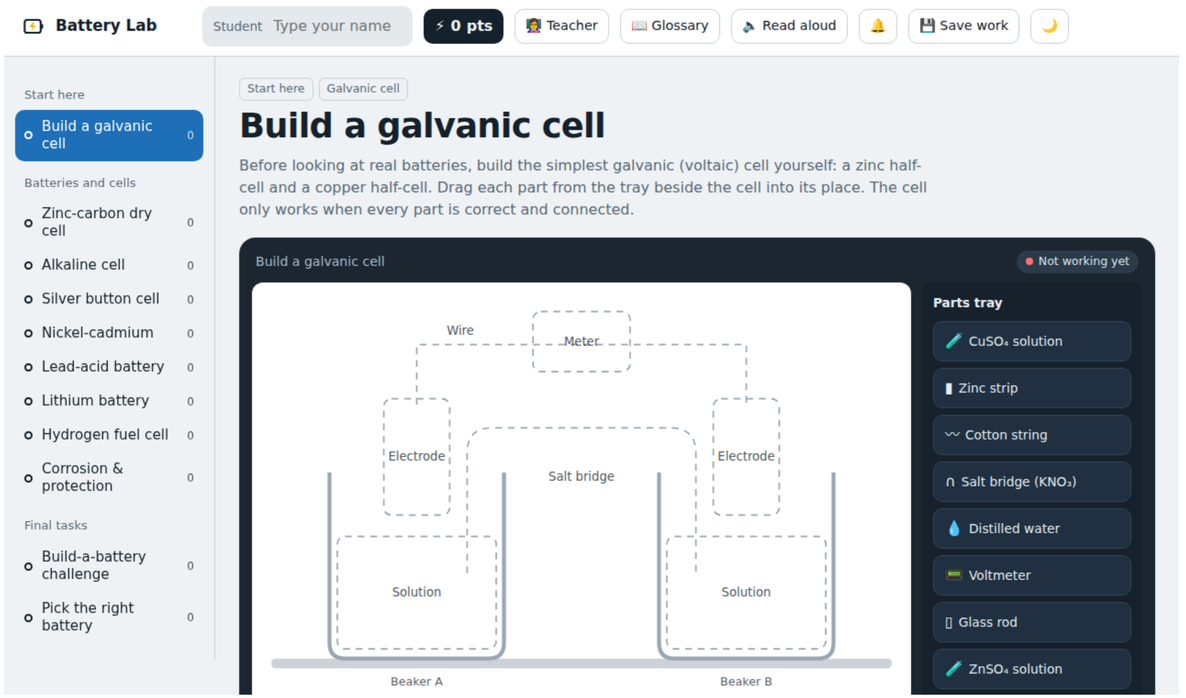
Battery Lab: Build a Galvanic Cell

Open the Battery Lab and build the zinc–copper cell yourself before we calculate its voltage.

Your task

1. Drag the two solutions into the beakers.
2. Add the matching electrode to each beaker.
3. Choose a salt bridge that really carries ions — test the wrong ones too.
4. Connect the wires and the voltmeter, then record the voltage.
5. Copy the cell notation the lab builds for you.

EXPLORE Battery Lab: Build a galvanic cell



Click the picture to open the lab

Reduction Potential and the Standard Hydrogen Electrode

- The tendency of a substance to gain electrons is its reduction potential.
- When two half-reactions are coupled, the voltage generated corresponds to the difference in potential between the half-reactions.
- **The standard hydrogen electrode** is a small sheet of platinum immersed in a hydrochloric acid solution with a hydrogen ion concentration of 1 M.
- Hydrogen gas (H_2), at a pressure of 1 atm, is bubbled in and the temperature is maintained at 25°C.

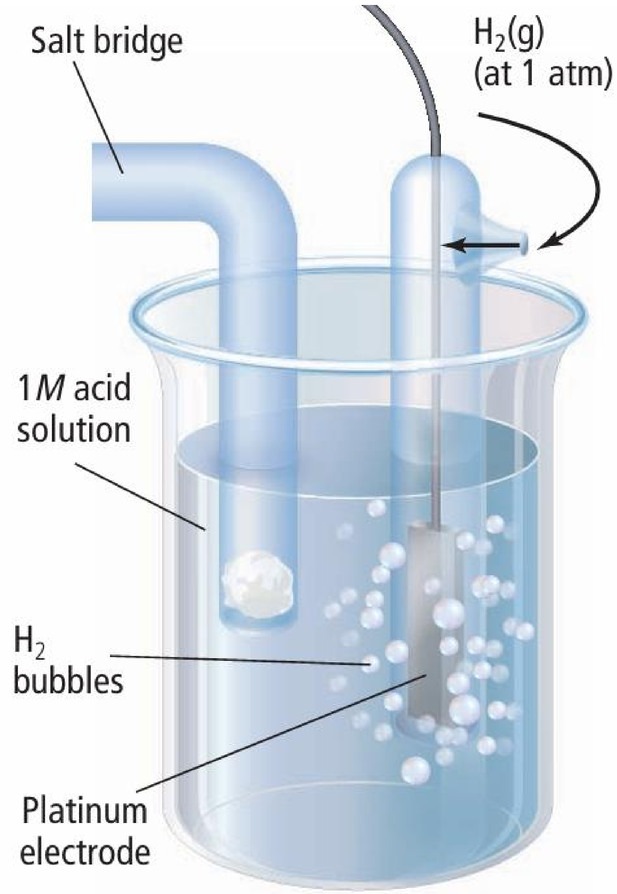
At the hydrogen electrode E° = 0.000 V

REDUCTION

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

OXIDATION

$$H_2(g) \rightarrow 2H^+(aq) + 2e^-$$



Standard hydrogen electrode

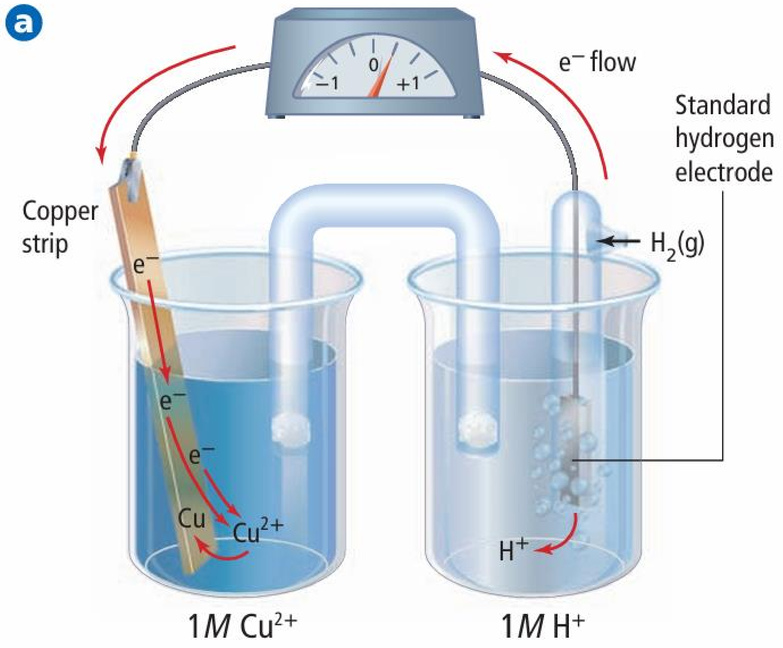
Table 1 — Selected Standard Reduction Potentials

The standard hydrogen electrode is the standard against which all other reduction potentials are measured (25°C, 1 atm, 1 M).

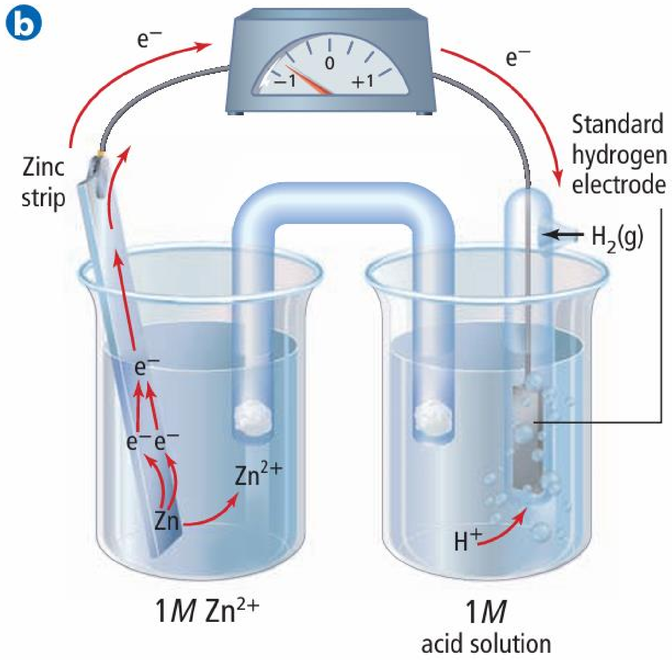
Half-reaction	E° (V)
$Li^+ + e^- \rightleftharpoons Li$	–3.0401
$Na^+ + e^- \rightleftharpoons Na$	–2.71
$Mg^{2+} + 2e^- \rightleftharpoons Mg$	–2.372
$Al^{3+} + 3e^- \rightleftharpoons Al$	–1.662
$Cr^{2+} + 2e^- \rightleftharpoons Cr$	–0.913
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	–0.7618
$Cr^{3+} + 3e^- \rightleftharpoons Cr$	–0.744
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	–0.447
$Co^{2+} + 2e^- \rightleftharpoons Co$	–0.28
$Sn^{2+} + 2e^- \rightleftharpoons Sn$	–0.1375

Half-reaction	E° (V)
$Pb^{2+} + 2e^- \rightleftharpoons Pb$	–0.1262
$2H^+ + 2e^- \rightleftharpoons H_2$	0.0000
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+0.3419
$I_2 + 2e^- \rightleftharpoons 2I^-$	+0.5355
$Ag^+ + e^- \rightleftharpoons Ag$	+0.7996
$Hg^{2+} + 2e^- \rightleftharpoons Hg$	+0.851
$Pt^{2+} + 2e^- \rightleftharpoons Pt$	+1.18
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.35827
$Au^{3+} + 3e^- \rightleftharpoons Au$	+1.498
$F_2 + 2e^- \rightleftharpoons 2F^-$	+2.866

Measuring Potentials Against the Hydrogen Electrode



(a) Copper is reduced — electrons flow to the copper strip



(b) Zinc is oxidized — electrons flow from the zinc strip

$H_2 \mid H^+ \parallel Cu^{2+} \mid Cu$

$E°_{Cu} = +0.342$ V

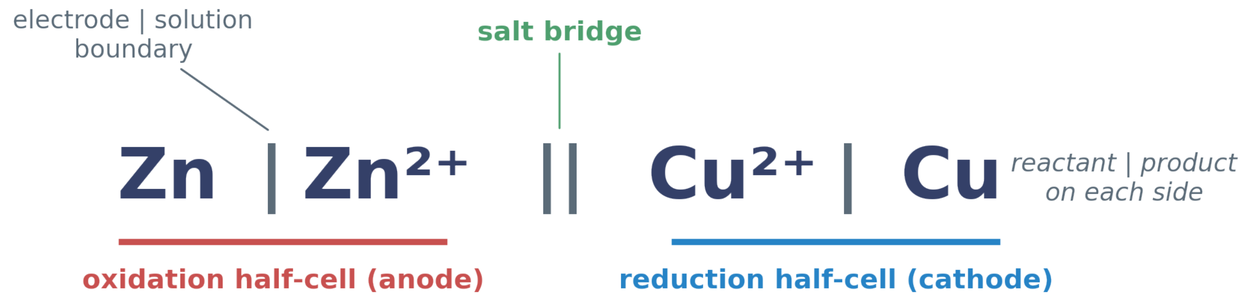
$Zn \mid Zn^{2+} \parallel H^+ \mid H_2$

$E°_{Zn} = -0.762$ V

Formula for Cell Potential

$$E°_{cell} = E°_{reduction} - E°_{oxidation}$$

standard potential of the reduction half-cell minus standard potential of the oxidation half-cell



Reading cell notation

What each symbol means

- **$E°_{cell}$** — overall standard cell potential
- **$E°_{reduction}$** — standard half-cell potential for the reduction (cathode)
- **$E°_{oxidation}$** — standard half-cell potential for the oxidation (anode)
- Use both values exactly as listed in Table 1 — do not change their signs.

Worked Example: Calculate a Cell Potential

The following reduction half-reactions represent the half-cells of a voltaic cell. Determine the overall cell reaction and the standard cell potential. Describe the cell using cell notation.

Given (Table 1)

$E° = +0.5355$ V

$I_2(s) + 2e^- \rightarrow 2I^-(aq)$

$E° = -0.447$ V

$Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$

Solve

1. **Reduction:** the half-reaction with the higher E° is the reduction — $I_2(s) + 2e^- \rightarrow 2I^-(aq)$
2. **Oxidation:** reverse the other one — $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$
3. **Overall:** $I_2(s) + Fe(s) \rightarrow Fe^{2+}(aq) + 2I^-(aq)$
4. **Potential:** $E°_{cell} = +0.5355\text{ V} - (-0.447\text{ V}) = +0.983\text{ V}$
5. **Notation:** $Fe \mid Fe^{2+} \parallel I_2 \mid I^-$
6. **Check:** $E°_{cell}$ is positive, so the reaction is spontaneous.

Practice: Cell Potentials

For each pair of half-reactions, write the balanced equation for the overall cell reaction and calculate the standard cell potential. Describe the reaction using cell notation.

$E^\circ_{cell} = E_{Red} - E_{oxi}$

$Pt^{2+} = 1.18$

$Sn^{2+} = -0.137$

PROBLEM 1

$Pt^{2+}(aq) + 2e^- \rightarrow Pt(s)$ and $Sn^{2+}(aq) + 2e^- \rightarrow Sn(s)$

$1.18 - (-0.137) = 1.3\,V$

$Sn|Sn^{2+} \,||\, Pt^{2+}|Pt$

-0.744

PROBLEM 2 -0.28

$Co^{2+}(aq) + 2e^- \rightarrow Co(s)$ and $Cr^{3+}(aq) + 3e^- \rightarrow Cr(s)$

$-0.28 - (-0.744) = 0.464$

PROBLEM 3

$Hg^{2+}(aq) + 2e^- \rightarrow Hg(l)$ and $Cr^{2+}(aq) + 2e^- \rightarrow Cr(s)$

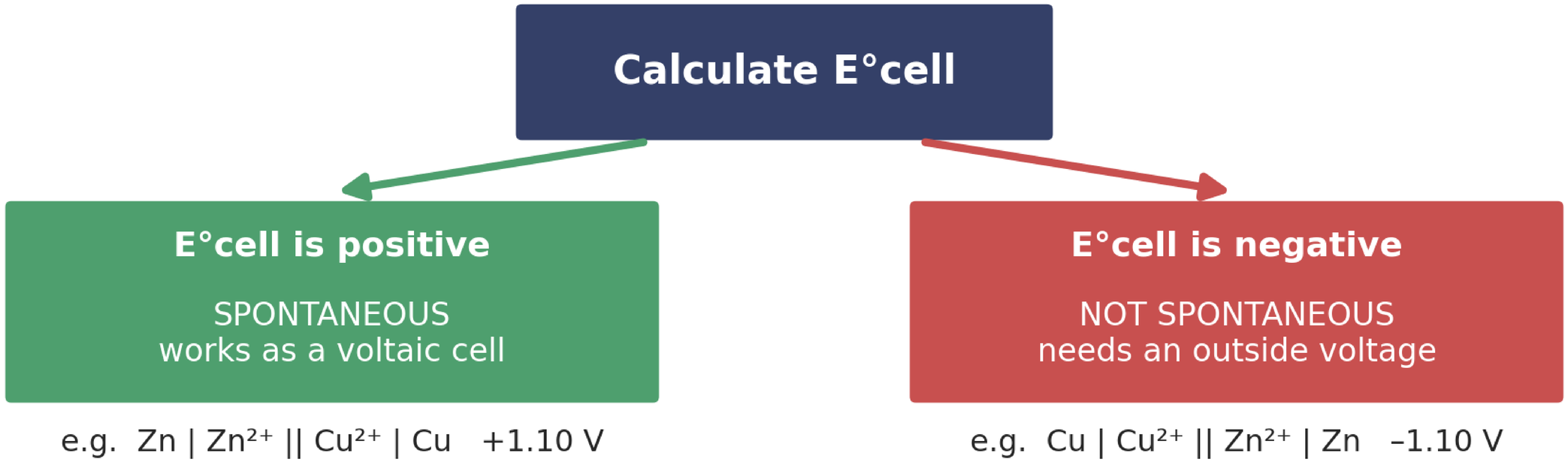
$0.851 - (-0.913) = 1.764$

Method

1. Look up both E° values in Table 1.
2. Higher E° = reduction (cathode).
3. Reverse the other half-reaction (anode).
4. Balance the electrons, then add.
5. E°cell = E°reduction – E°oxidation

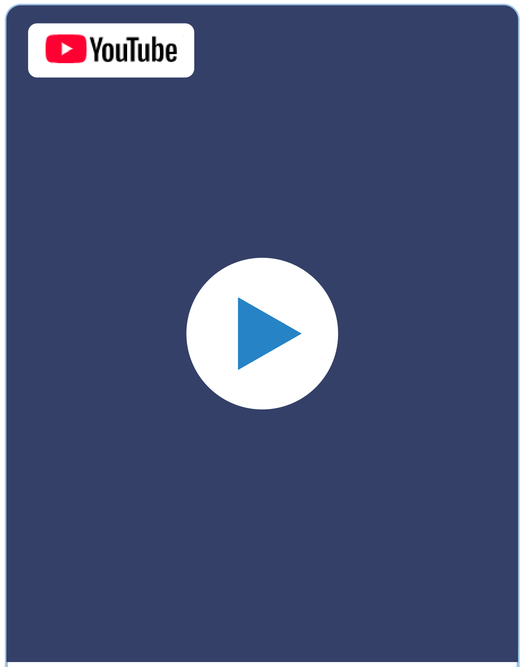
Using Cell Potentials: Will It Happen?

- Cell potentials can be used to determine if a proposed reaction under standard conditions will be spontaneous.
- If the calculated potential is positive, the reaction is spontaneous. If it is negative, the reaction is not spontaneous.



Watch & Explore: Voltaic Cells

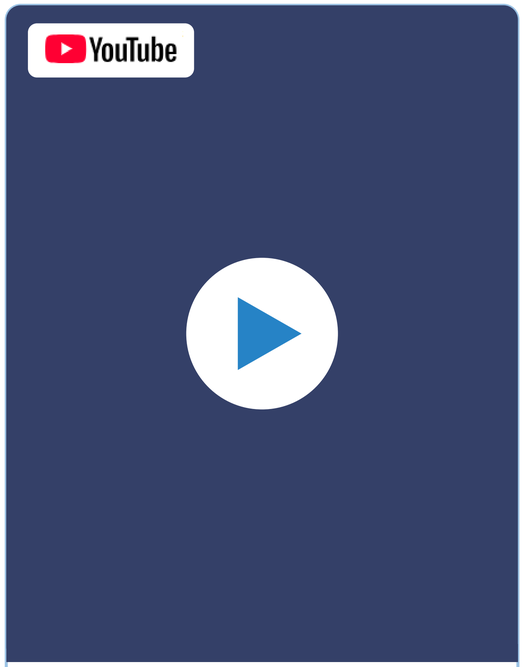
Click a card to open it. Build at least two different cells and compare the measured voltage with your calculated E°cell.



Galvanic Cells (Voltaic Cells)

Tyler DeWitt • 23 min

Anode, cathode, half-reactions and electron flow.



Introduction to Galvanic Cells & Voltaic Cells

YouTube • 28 min

A second walk-through with cell potential examples.



Galvanic/Voltaic Cells

AACT simulation

Choose metals and solutions; watch the particles.



Battery Lab: Build a galvanic cell

Class lab (HTML file)

Drag the parts into place until the cell works.

Checkpoint: Voltaic Cells

1. At which electrode does oxidation occur? At which does reduction occur?

2. What would happen to the current if the salt bridge were removed? Why?

3. In the zinc–copper cell, in which direction do electrons flow through the wire?

4. What is the standard reduction potential of the standard hydrogen electrode?

5. A calculated E°cell is –0.46 V. Is the reaction spontaneous as written?

Quick check — discuss with a partner, then confirm with your teacher.

Lesson Summary

Key Takeaways

- In a voltaic cell, oxidation and reduction take place at electrodes separated from each other.
- The standard potential of a half-cell reaction is its voltage when paired with a standard hydrogen electrode under standard conditions.
- The reduction potential of a half-cell is negative if it undergoes oxidation when connected to a standard hydrogen electrode, and positive if it undergoes reduction.
- The standard potential of a voltaic cell is the difference between the standard reduction potentials of the half-cell reactions.

Key Ideas & Equations

- **Anode** — where oxidation occurs
- **Cathode** — where reduction occurs
- **Electrons** flow from anode to cathode through the wire
- **Salt bridge** keeps ions from building up
- **E°cell** = E°reduction – E°oxidation
- **Positive E°cell** — spontaneous; negative — not spontaneous

Evaluation: Voltaic Cells

Complete independently. Review the summary if you get stuck, then check your answers with your teacher.

1. Describe the conditions under which a redox reaction causes an electric current to flow through a wire.

2. Identify the components of a voltaic cell and explain the role of each component.

3. Write the balanced equation for the spontaneous cell reaction between $Ag^+(aq) + e^- \rightarrow Ag(s)$ and $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$, and calculate E°cell.

4. Under what conditions are standard reduction potentials measured?

5. **Calculate:** E°cell for the cell $Zn \mid Zn^{2+} \parallel Ag^+ \mid Ag$, and state whether the reaction is spontaneous.

6. **Challenge:** Using Table 1, choose the two half-cells that would give the largest possible cell voltage. Write the cell notation, calculate E°cell, and suggest one practical reason this cell is not sold as a battery.

Self-Evaluation Rubric

4 – Confident mastery • 3 – Solid grasp, minor gaps • 2 – Needs review • 1 – Revisit the lesson